Track 2

Problem Statement

A Rapid DNA-Based Spot Detection Kit for Bovine/Animal Samples

Explanation

Species mislabeling and adulteration of meat and animal-derived products remain a persistent challenge for food safety, halal/religious compliance, and consumer trust. Conventional confirmation methods such as real-time PCR and DNA sequencing are highly accurate but require trained personnel, laboratory infrastructure, and turnaround times of several hours to days, making them impractical for on-site or field-level screening at markets, checkpoints, processing units, or border inspection points. Recent research has demonstrated that isothermal amplification techniques, such as recombinase polymerase amplification (RPA), combined with lateral flow strip detection, can identify species-specific DNA sequences within 20-35 minutes with results visible to the naked eye, without the need for a thermocycler. In India, institutions such as the ICAR-National Meat Research Institute's Meat Species Identification Laboratory already provide NABL-accredited molecular testing for species authentication, but such laboratory capacity is limited relative to the scale of the informal meat trade. There is a clear need for a low-cost, field-deployable, DNA-based spot detection kit that can rapidly confirm or flag the presence of specific animal-origin DNA (e.g., bovine, buffalo, porcine, or other species) directly at the point of sampling, supporting faster enforcement, quality control, and consumer protection while remaining consistent with applicable food-safety and regulatory frameworks.

Objectives

- To enable rapid, on-site confirmation of animal-origin DNA in raw, processed, or ambiguous meat/animal-derived samples without laboratory dependency.
- To reduce the turnaround time for species identification from hours/days (standard PCR/sequencing) to under 30-40 minutes.
- To support food safety officers, market inspectors, and quality-control personnel in detecting adulteration, mislabeling, or unauthorized species substitution.
- To provide a low-cost, easy-to-use alternative to laboratory-based molecular testing that requires minimal training and no specialized equipment.
- To ensure results are interpretable, reliable, and consistent with applicable food-safety, halal-certification, and regulatory testing frameworks.

Expected Features

The proposed kit/platform may include the following capabilities:

1. Rapid DNA Extraction

A simplified, minimal-equipment DNA/nucleic-acid extraction protocol (e.g., lysis-buffer-based or bead/stick-based) that can be completed within 1-5 minutes at ambient temperature.

2. Isothermal DNA Amplification

Use of isothermal amplification methods (e.g., recombinase polymerase amplification or loop-mediated isothermal amplification) that avoid the need for a thermocycler and can run at a single, near-room temperature.

3. Species-Specific Target Design

Primers/probes targeting conserved, species-specific mitochondrial regions (e.g., cytochrome b, COX genes) validated for high specificity against non-target species and low cross-reactivity.

4. Lateral Flow Visual Readout

A strip-based lateral flow detection format that produces a visible colored line (naked-eye readout) within minutes of amplification, requiring no specialized reader for a basic positive/negative result.

5. Multiplexed Detection

Ability to screen for more than one target species in a single test (e.g., differentiate bovine/buffalo from porcine or other species) using distinguishable line positions or colors.

6. Field-Deployable Kit Format

A compact, portable kit design suitable for use by inspectors, veterinary officers, or field staff at markets, processing units, and checkpoints without laboratory infrastructure.

7. Digital Result Logging (Optional Companion App)

An optional mobile application to photograph/interpret strip results, timestamp and geotag tests, and maintain a digital record for compliance and audit purposes.

8. Sensitivity & Specificity Validation

Documented limit-of-detection and cross-reactivity testing against a panel of common target and non-target species to support the kit's reliability claims.

9. Cold-Chain-Independent Stability

Reagents formulated or lyophilized for stability at ambient or minimally refrigerated conditions, supporting use in remote or resource-limited settings.

10. Scalability & Cost-Effectiveness

A design and manufacturing approach that keeps per-test cost low enough for routine, high-volume field screening, with the ability to extend the panel to additional species over time.

Suggested Tech Stack

Participants are encouraged to justify their technology and assay-design choices based on sensitivity, specificity, field usability, cost, and stability rather than adopting any predefined method.

Resources

[1] Indian Council of Agricultural Research (ICAR), National Meat Research Institute (NMRI), "Meat Species Identification Laboratory: A Beacon of Integrity in Meat Authentication and Food Fraud Investigation," ICAR-NMRI, Hyderabad, India. [Online]. Available: https://icar.org.in/ (or the specific ICAR-NMRI webpage). [Accessed: Jul. 17, 2026].

[2] Food Safety and Standards Authority of India (FSSAI), "Regulatory Framework Overview," Drishti IAS. [Online]. Available: https://www.drishtiias.com/ (article on FSSAI Regulatory Framework). [Accessed: Jul. 17, 2026].

[3] X. Zhang, Y. Wang, H. Liu, *et al*., "Lateral Flow Biosensor Utilizing Self-Assembled DNA-Multicolor Au/Ag Nanoassemblies and Dual-Tailed Recombinase Polymerase Amplification for On-Site Triplex Detection of Chicken, Duck and Pork Adulteration in Beef," *Food Chemistry*, Elsevier, 2025. [Online]. Available: https://www.sciencedirect.com/. [Accessed: Jul. 17, 2026].